

A Mythic Economy

Designing a quasi-ancient society for Mythras and other RPGs

by Jonathan Drake

A fantasy setting is usually a world that bears some resemblance to the real world of past ages. If roleplaying games began with the shared language of a pseudo-medieval setting, over the years almost every age of history has been plundered for inspiration, and sometimes 'realistic' detail, to flesh out a fantasy. You don't have to think too hard about what makes the world in which you set your adventures hang together – it's entirely possible to focus just on the story and action and skate over the rest. On the other hand, a basic grasp of how its economy works can be really useful. It answers the question, "how do people live?" When an adventure is over, it's always good to have an idea about what the adventurers can do with their treasure. Most of all it explains what the vast majority of people in your world spend their time doing and what their priorities are, including the friends, relatives and adversaries your adventurers interact with. This, in short, is getting enough to eat to stay alive and healthy, and for many people that is quite enough of a challenge even before the marauding hordes of Chaos arrive in the agricultural backwater they call home.

What follows lays out an 'ancient' rather than a 'medieval' model for a historical or fantasy setting. Many years ago White Dwarf magazine published a pair of articles on the subject of 'Designing a Quasi-Medieval Society' for D&D. It's entirely appropriate that an article about designing an 'Ancient' society should, at least in principle, be targeted at the RQ/BRP family of games in general, since it was the

original RuneQuest game that took RPGs into worlds with an ancient feel for the first time.

The technologies available are more or less the same, but the way people organise themselves is very different.

The information given here assumes a number of things:

- 1) Agriculture in your campaign world makes use of similar crops to real world examples from pre-modern Europe and the Mediterranean and achieves similar yields.
- 2) The food supply includes a wide variety of different grains, fruit, vegetables, meats, fish etc, however some form of grain provides the vast majority of all calories consumed.
- 3) Human beings are the same as in the real world – they have standard Characteristics and Attributes, require the same amount of sustenance, and are capable of the same amount of physical labour.
- 4) The year in your setting is approximately 360 days long and contains 12 months. A single harvest per year for each crop or field is the norm but there may be exceptional circumstances that allow more than one to be achieved.

CURRENCY

CONVENTIONS

I'll be using the standard 1SP = 10CP throughout this article. It works well, and here's why:

The Bread Standard

A typical peasant or poor townsman consumes enough calories (2,000 – 3,500, depending on age, gender and profession) to sustain him through a day of physical work, usually in the form of bread (2 x1lb – 0.5kg loaves), or some sort of porridge. A happy citizen is one who knows that they can afford to eat, and for those living in towns that means being able to buy bread. The price of a loaf of bread is an essential indication of how society is getting on. If you can't buy one with your smallest coin (1CP), it may be time for a food riot, at which point the government usually has to intervene in the economy.

The Grain Standard

To ensure you have your daily bread you need a secure supply of grain – whether produced locally or shipped in from elsewhere. The standard measure in which grain is bought and sold is not the single loaf (or the grain to make it), but the "peck". This is 6.67kg, a small sack, just under 15lbs of dry measure. Incidentally, there are four pecks to a bushel - unless you are a Roman, when there are three, or a Greek, when there are six. The point is, the peck is a constant measure, and grain is the ubiquitous commodity of an ancient economy, so it can act as our standard unit of value. By comparing the price of one peck of grain to the price of other goods and services in the same historical time period, we can calibrate a price list. We'll peg

the price of 1 peck of grain at 1SP. As it can produce 16-20 loaves of bread, its end product is worth up to 2SP. The difference between the two is accounted for by storage, transport, milling, and baking, plus any loss or spoilage along the way. This is a reasonable premise, and is borne out by the relative costs of grain and bread in the ancient sources.

CITY STATES

Antiquity is a world of cities, each one created when a bunch of connected villages and towns combine their political, religious, market and military institutions, and choose a single settlement to house them all - a process known as *synoikism*. A 'city-state' therefore consists of an urban centre (some of them really no bigger than large towns, some of them a teeming metropolis) and its territory, which will of course contain many villages, hamlets and estates – but rarely isolated homesteads. The ideal state is economically self-sufficient and politically independent. Its citizens participate in its politics and fight in its armies. Even if conquered and no longer independent, the urban centre still acts as the administrative, religious and market centre of its region.

How Big Is the City?

A city can come in many shapes and sizes. At the bottom end, it must be big enough to be capable of fielding a creditable force for self defence. This may be just a few hundred warriors, with a total population of perhaps 2,000 - about right for a newly founded colony or small island state. There is no upper limit, however the available land around each city will limit its growth, and once the capacity of its territory is exceeded it can grow no more

Total Population	10,000	30,000	100,000	250,000	500,000
Urban Population (20%)	2,000	5,000	20,000	50,000	100,000
Rural Population (80%)	8,000	25,000	80,000	200,000	400,000
Total Arable Requirement	3,600 Hectares	10,800 Hectares	36,000 Hectares	90,000 Hectares	180,000 Hectares
Minimum City Territory	120 km ²	360 km ²	1200 km ²	3,000 km ²	6,000 km ²

unless it can ship in food from elsewhere or simply take new land from other people.

As a rule of thumb assume that no more than 30% of a city's territory can be dedicated to intensive food production, at an average net yield of 100 pecks of grain per Hectare. We need something in the region of 240kg of grain per year (36 pecks = 2,300 calories per day) or equivalent to feed each inhabitant. As a result you would expect a self-sufficient city's total territory to be 120km² for every 10,000 inhabitants. A culture that practices alternative year fallowing needs double that, crop rotation perhaps +50%.

If a city does not have access to this much land, or is situated in a region where cultivating 30% of the land surface is unrealistic (e.g. due to the extent of desert, steep hills, mountains, forest or water), then you should assume it is importing food from elsewhere, and think about how it covers the cost of that. If for some reason that's not possible, then your city may be too populous or there must be some other circumstances - whether a magical effect that explains its ability to survive or the presence of some super-fertile soils that guarantee a higher than average crop yield (such as the Nile valley). If your city is located amid rolling plains with much more than 30% of its land ideal for grain cultivation, you can assume it is a net exporter and can grow rich on the proceeds.

Non-Arable Land

The city's territory that is not ploughed for planting grain has its uses. This will include forestry, not just for timber but also for harvesting nuts, berries and fruit as well as for hunting; coastlines, rivers streams and lakes that yield fish; pastures, hillsides that are too steep for ploughing but that may suit vines very well, or just land given over to cash crops such as cotton or flax, or even growing animal fodder. Unproductive land may still have its value - clay pits for brick-making, mining for salt or metals. All these things provide the citizens with the means to live beyond

subsistence, and provide opportunities for a wide range of crafts and the support of specialists, most of whom will be concentrated at the city itself.

WAGES AND COST OF LIVING

Once you take into account that, before the modern era, food should account for some half of all expenditure, 4CP per day is more or less the functional minimum wage. Therefore perhaps 150 - 200SP per year is a good living for a young man without dependents to support, but it is nowhere near enough if you have mouths to feed besides your own. It is assumed that the head of the household accounts for 40% of food and other expenditure. Hence an income of 360SP or more is a reasonable estimate of target income for an average family of two adults and 2-3 children.

For most people this is not easily achieved without everyone in the family who is old enough to do so making themselves economically useful. The suggested pay and prices presented in this section are based on known ancient examples, and using the grain standard of 1 peck = 1SP = 10CP to benchmark them.

Job	Daily	Annual	Monthly
Craftsman/Skilled Worker (high)	1SP, 3CP	336SP	28SP
Low-Skilled Craftsman or Labourer	6CP	144SP	12SP
Elite Artisan (e.g. Artist)	4SP	1080SP	90SP
Secretary	1SP, 1CP	288SP	24SP
Teacher*	9CP	240SP	20SP
Messenger	7CP	180SP	15SP
Priest, Soothsayer, Temple attendant*	7CP	192SP	16SP
Soldier (Professional)*	1SP	240SP	20SP
Soldier (Militia)*	5CP	182SP	15SP
Army Officer	22SP	5760SP	480SP

* These jobs typically get basic maintenance covered, so the payment represents "take home pay".

The daily rate is higher than the annual income would suggest – this is because people don't work every day – so it has been adjusted accordingly. Workers generally put in about 260 days per year - the remainder being one sort of holiday or festival or another - which also applies to slaves, who are (at least in theory) way too expensive and valuable to kill through overwork.

You can see that military rates of pay tend to be high. The income shown here is also expected to be boosted with cash bonuses, booty, and even a grant of land at the end of service. This is because professional soldiers can cause a lot of trouble if they get fed up with their pay and ordinary citizens need to see an upside to taking time out to serve in the militia.

A skilled worker or artisan earning 336SP per year, with a wife who devotes half her time to her family and the remainder to earning the equivalent of a labourer's or unskilled wage of 144SP through piece-work, spinning etc can between them raise a family on a combined 480SP.

Working the Land

Most people do not earn regular wages but work their own plot of land. Assume that somewhere between 70% and 90% of all adults are working in agriculture.

Each Hectare under grain should on average deliver a net 100 pecks at harvest time, which equates to some 100SP of earnings for the farmer. It follows that the right size plot to support a family is about 4 Hectares. You need a bit more than this if you practice fallowing and crop rotation, but as it happens 8 Hectares is regarded as the maximum that one man (with help from immediate family), can successfully farm.

For most people a holding that size is something they can only dream of. Many property-owning citizens make do with only half a hectare. With a plot that small a farmer can just about subsist but he cannot support a family – he will need to find additional sources of income or pay. Very often this is how the state, or the rich, like things to be. It means the poor have something to call their own, but also are dependent on hand outs, employment and loans from their social superiors to make ends meet. These are all opportunities to create loyal dependents, recruits for the army, and labour for the construction of public works or the estates of the wealthy. If however things become too exploitative, at some point the poor (who are also by definition “the many”) get angry, agitate, riot and rebel, so there is a balance to strike.

THE PRICE OF GOODS

Some basic items can cost a considerable proportion of your earnings. On the whole the prices given below provide a reasonable approximation for a ‘realistic’ price model using patchy ancient data. Clothing in particular is an expensive necessity; hence the poorest folk and slaves might have no more than a simple shift of unbleached wool to their name, or even something sewn together from scraps. It also explains why people in antiquity got so

exercised about what we might consider petty thefts (such as having your cloak stolen at the baths).

Clothing and Leather Goods	
Basic tunic or shift of unbleached wool, for a slave or peasant	2SP
Simple Tunic	4SP
Tailored and Dyed Tunic	7SP
Tailored, Dyed and Decorated Tunic	15SP
Basic Cloak	1SP 2CP
Basic Shoes or Sandals	8CP
High Status Shoes	1SP 5CP
Military or Working Boots	1SP 2CP

Armour and Weapons

One area where the pricing given in game rules can appears out of touch with realistic pricing is in weapons and armour. To some extent this is because this is the stuff adventurers are looking to upgrade all the time, and pricing them high makes for logical progression as characters become more skilled, better equipped, and take on more dangerous challenges. A Greek hoplite panoply of spear and shield is reckoned to have cost around a month's average wages - and adding the simplest form of helmet and leather or cloth armour not much more than that.

Armour and Weapons	
Simple Metal Helmet	5SP
Spear	2SP
Hoplite Shield	20SP
Sword	25SP - 50SP
Leather Armour	5SP per location
Bronze Armour	20SP per location

Livestock

Slaves and Livestock	
Agricultural Slave	200SP - 400SP
Domestic Slave	500SP
Premium Slave	1000SP–2,000SP
Donkey	120SP
Riding Horse	200SP - 400SP

If animals are to be used for food, their price will increase as they get closer to slaughtering age and weight. Up until that point they still need to be pastured and fed – land and fodder cost money, and there's the ongoing risk of loss to disease, predators and thieves. Younger live animals (slaves included) are cheaper as the buyer has to spend money on feeding them until they can be made full use of.

For real world creatures the most efficient calorie-provider is the humble pig. A young 60kg pig should on average be worth 20SP in meat, fat and offal and one that has been allowed to reach a mature weight of over 130kg more than twice that.

THE PRICE OF LAND

When an Adventurer returns from an expedition with a big pile of treasure, one of the most natural things to do is to use it as capital to buy land. An average price for land is 1,000SP per Hectare. The financial reason for buying land is to put your money into something that should pay dividends every year through the produce of the fields. It is much the preferred form of investment, partly because other ventures, such as trade, come with significant risks attached. If you do want to raise funds for a risky venture, having land allows you to offer collateral for a loan.

Ownership of land is so important that it is normal to deny citizenship to those who don't have any, and to deny ownership of land to people who are not local-born citizens. There may be royal and temple estates, and sometimes the state makes rules about who can own what, but the vast majority of land is in private hands and may be passed on through inheritance, bought, sold and rented out.

Profiting from the Land

1 Hectare of land requires 20 pecks of grain to seed it, and a farmer would normally hope to get six times as much grain from the harvest as he planted. Of this he needs to set aside 20

pecks for the next year's planting. Different soils and climates will result in different yields, and this has a knock-on effect on the value of the land and how much of it you need to maintain your status

A very small plot may be farmed without a plough, but anything substantial will require a pair of oxen and a ploughshare - larger farms will have their own, smaller ones will share or borrow as they can. If you have a larger plot of land, you need more people to work it. Renting out is popular because although it delivers less return it requires no capital outlay on slaves, and the tenants are expected to join your gang of supporters. A typical rent would be between 15% and 20% of the tenant's produce.

Slaves are expensive at start-up, but more profitable in the long run. An agricultural slave costs 200-400SP, but doesn't come with a family to feed unless you also invest in breeding stock. In this case you get all the benefit of your additional Hectares, less perhaps 100SP per annum for maintenance (including food, some basic clothing and pocket money). Immediately after a war slaves might be available cheaply, so that can be a good time to retire to the country life.

Inheritance

An Adventurer may be in line to inherit a plot of land – however in the real world

primogeniture, in which the eldest son inherits his father's lands and title by default, is a medieval invention. In an ancient economy property is normally divided up between the surviving sons and possibly daughters – or anyone else that the deceased has nominated in his will. This can of course lead to small farms being partitioned into uneconomically small units. It also leads to a great deal of intrigue, plot and politics over the inheritance of large estates.

ADVENTURER LIFESTYLES

This table takes a broad view of the money needed for an individual Adventurer to enjoy a certain lifestyle. This might be useful in calculating cost of downtime, but also there will be occasions where an Adventurer might spend a lot of money for a short period of time to give the impression he is richer than he is. If the Adventurer has a family to feed, multiply these living costs by up to 2.5.

A suggestion is also made here regarding starting money if you are using the price guides in this article, given that the standard 4D6x75SP might prove inappropriate for a pauper on the bread line.

Lifestyle	Weekly Cost	Monthly Cost	Annual Cost	Starting Money
Pauper	14CP	6SP	72SP	D6SP (average 35CP)
Poor Townsman or Peasant	3SP	12SP	144SP	2D6x10SP (average 70SP)
Yeoman Farmer or Artisan	4SP	16SP	200SP	2D6x25SP (average 175SP)
Well-to-do Merchant or Businessman	10SP	40SP	500SP	4D6x75SP (average 1050SP)
Gentleman	10SP	40SP	500SP	4D6x75SP (average 1050SP)
Rich Man	20SP	80SP	1,000SP	6D6x100SP (average 2100SP)
Aristocrat or Land Magnate	100SP	400SP	5,000SP	2D6x1,000SP (average 7000SP)

Soil Quality	Yield	Pecks (Per Hectare)	Price of Land (SP/Hectare)	Subsistence Holding (Hectares)	Yeoman Holding (Hectares)	Gentry Holding (Hectares)	Aristocratic Estate (Hectares)
Poor	x4	80	600	2	NA	40+	400+
Average	x6	120	1000	1.5	8	20+	200+
Good	x8	160	1400	1	6+	15+	150+
Fine	x10	200	1800	0.75	5+	12+	120+
Rich	x12	240	2200	0.66	4+	10+	100+

THE BREAD LINE

Some people live close to destitution. They are most likely to be young, old and infirm lacking someone to support them, and unable to offer up the strength of their bodies in return for pay. A city dweller who has to pay for baked bread needs 2 CP per day, 70SP per year, or goes hungry. Sometimes a simple porridge can provide a cheaper alternative, although that might lead to a daily hunt for something to make it more palatable. A lean-to, shack or hovel may be the pauper's only shelter, and for clothes he has only rags cast off by someone else. People living in such dire straits are a minority, but they are enough to be visible. Whenever drought, famine or war strike significant numbers of people are easily tipped into the same boat and driven to extreme behaviours such as infanticide (to reduce the number of dependent mouths), selling themselves or their children into slavery, begging or taking up crime.

Historically some 65% of people lived in an economic state we would consider poor in absolute terms – meaning that they lived with continual uncertainty about how they would survive year by year. Consequently if someone of means could remove uncertainty from such people by ensuring food supplies, giving out a dole, providing employment or whatever, they could earn the gratitude, loyalty and service of a lot of people. Most ancient states left individuals to build these webs of influence through generosity. A democratic state might start introducing state hand-outs, payments for civic or military service and the like, thereby

liberating the poor. The poorest citizens may be able to get state relief if they are a citizen, usually in the form of some sort of food handout, but potentially a new start in life when land becomes available to resettle them as a result of commandeering land from the rich, or by foreign conquest.

Tenant Farmers and Debt-bondsmen

One alternative to destitution is to find a landlord who will rent you a piece of his land, taking a share of the produce as payment. In Archaic Athens (roughly 750-500 BC in traditional reckoning), these farmers paid a sixth of their produce in rent to the landholder. Clearly there were many who had small or unproductive plots, or who just had a bad year, and having nothing to spare after feeding their families fell into arrears. The only thing they had on which they could secure the debt was their own bodies – so inevitably many fell into slavery as “debt-bondsman”. The effect was simmering unrest, until land was redistributed to the poor to free them from dependency. More viable plots can be rented, and in this case the tenant may have plenty of margin from which to pay the rent; however the majority of tenant farmers are close to the bread line.

The Rich

The purpose of wealth is not to work and pursue the good life, whether that means a life of leisure or doing something useful like practising law, literature, politics or war, unhindered by the need to spend time on base concerns like industry and commerce.

Managing a large estate worked by slaves through a steward will generate significantly more income than renting your land out to tenants, but that approach demands a good supply of slaves, and also fails to deliver the important benefit of tenant farmers who owe you loyalty and service in addition to rent. In general the more freedom, land ownership and economic self sufficiency is enjoyed by ordinary citizens, the less inclined they are to do what the upper classes ask of them and the more the rich turn to slave labour.

In any normal city state those who count themselves the rich are probably generating some 3,000 pecks or more from their land every year in rents, based on taking a share of 15%+ from the available product. If you are using the Empires supplement this usefully equates with the average income in SP of a Knight. In classical Athens an income of 500 Greek *medimnoi* (bushels) was sufficient to qualify for the highest rank of citizen, and 500 Athenian bushels are the same as 3000 pecks. Similarly in Rome you needed to own property worth 1 million Sesterces to be a Senator, which is equivalent to holding 250 Hectares of decent agricultural land and which produces much the same result. This is of course only entry-level rich and a state of any size will have individuals who have holdings worth many times this amount. The effect of conquest and Empire may be to massively increase how rich it is possible to be. Individuals may accumulate holdings running into thousands of Hectares.

GOVERNMENT and TAXES

In an ancient economy taxes are not, after all, a certainty of life. There is generally no such thing as an income tax, and often no direct taxation of a state's citizens at all, but it is quite possible resident foreigners are subject to a poll tax.

The most common tax is on trade - the state raises money from duties on both import and

export, at a typical rate of 5%. Any goods not declared to port officials are liable for confiscation.

A Sales Tax is common, the percentages applied normally quite low – perhaps as low as 0.5% and no higher than 3%. A farmer selling his own produce might be exempt – this is really there to catch traders who buy from one person to sell to another at a mark-up without having actually produced anything.

A major resource – such as a salt works or silver mine - would be heavily taxed, if not actually taken over by the state and exploited directly. The richest silver mines in the ancient world might yield millions of Silver Pieces in value every year. These resources are definitely worth fighting for – your setting may have one or two mines that can match the entire output of a major city's agricultural production in value.

A one-off property or wealth tax can be ordered from time to time to raise funds from the citizenry to meet a crisis (particularly in war), but when the crisis has passed there may even be efforts to repay the money - showing it is a sort of unsecured, obligatory loan to the state rather than a tax.

Foreign possessions are a different case, and empires can bring dividends. Foreign subjects have their land and property assessed for tax – typically at 1% of value, and the money is collected either by their own local authorities and handed over as tribute, or collected by a private firm of 'tax farmers'. These people bid money for the right to collect tax in a given region or city and make a margin on the difference between what they bid and the amount actually raised.

State Expenditure

Most public works are paid for not from the public purse but by the rich, directly from their own resources. This is sometimes voluntary, sometimes highly regulated, but always required. An aristocrat who pays for construction of a temple or market place, or to

outfit a warship, or for a public entertainment or religious festival, gets to show how much richer and public spirited he is than his peers. His generosity is a means of competing for the loyalty and votes of ordinary citizens.

The state treasury needs to be able to cover the cost of wars, and also to ensure there are reserves of both food and money to plug the gap in the event of some disaster. If there is plenty of surplus it might be spent on temple-building, or the sort of public works that are beyond the resources of the richest citizen. A city tends not to spend its money on a civil service. Elected magistrates at all levels do their job in peace and war as amateurs. They probably do so at their own cost, a system which both 'justifies' profiteering or embezzlement and of course puts public office beyond the reach of anyone who is not very rich or under the patronage of someone who is.

Money

Every city likes to mint its own coins. Coins are in principle valued according to the weight of precious metal they contain. As a result they can be used abroad because a local money changer will be happy to convert to local currency for a small commission. However once the state gets into difficulties – not enough bullion for the number of coins they plan to mint, for example – they may start to debase the coinage – that is, mix the content of the coin using cheaper metals. When this happens the state's own credit rating is paramount. The coinage is now only worth what it says it is if the issuing authorities can be relied upon to make good upon its face value. So if your adventurers stumble upon a big hoard of silver pieces, it may be those silver pieces are not pure – or legal tender. And what they think is a massive pile of cash is worth a fraction of what they believe. Conversely, they may come across some hoard of ancient coins that are in fact almost pure bullion, and the reverse may apply.

In Legend, a useful Common Magic spell would be one that tests the purity of a coin, or

weighs a product or substance without access to a scale.

Economic Wars

Access to good land is a key objective in any conflict; and that one of the best ways to hurt your enemy is to destroy the crops in the fields before the harvest. Another is to directly attack the enemy's cults or priesthoods that provide agricultural magic, or even to attack the very gods and spirits that give those cults their power.

In a war of conquest the hope of the poor, possessed of only the smallest holdings or no holding at all, is that they would be settled on land taken from other people and consequently become better off. For this very reason the ordinary people who had to fill the ranks of a citizen army regularly voted in favour of wars - in many places and times a declaration of war was the only decision in which the citizenry as a whole had the final say through an open vote. They might not only get a little pay and booty, but perhaps also the award of a much bigger parcel of land. If that results in farm a plot too big to exploit with just the labour of the farmer and his immediate family the original owners are subjected to them as tenants, or else prisoners of war subjected to slavery and put to work.

TRANSPORT and SHIPPING

Transporting goods by land is vastly more expensive than doing so by sea. Cities of any size are almost always on or close to the sea in order to make shipping commodities and other goods in and out a viable process. Ancient land transport is more expensive than during medieval times, apparently because people were limited to oxen for drawing wagons, having not yet invented a horse collar allowing a seriously heavy load to be pulled. The following costs give an idea of how much an Adventurer will have to pay to ship a big load from one place to another - or alternatively, the

sort of money he can charge for doing so himself. These costs include all hands required to do the job, as well as the use of the ship, wagon or beast of burden.

By Road

To shift 1 wagon load, or 500 ENC of goods (using 1 ENC = circa 1kg this is equivalent to about 75 pecks of grain) by ox cart will cost about 40SP per 100 miles; by camel, a load half that size will cost 16SP per 100 miles. A donkey will carry a load of up to 100 ENC for 8SP per 100 miles.

By Sea

Shipping by sea is much cheaper, at an average of 3SP per 1,000 ENC every 100 miles. Hence a full load of grain on a 250-ton transport ship costs 750SP per 100 miles – which is roughly per day of sailing. The cargo itself is worth 37,500SP.

By River

River traffic is slightly more expensive than seaborne traffic. Assume a cost of 10SP per 1,000 ENC per 100 miles

Transport	Load per Vessel, Wagon or Animal	Cost per Ton Per 100 Miles Transported
Sea	25-500 Tons	3SP
River	1-100 Tons	15SP
Ox Cart	0.5 Tons	80SP
Camel	0.25 Tons	64SP
Donkey or Mule	0.1 Tons	80SP